The Prime Lattice as a Prototype for the BSD Hamiltonian: Rank as Spectral Multiplicity and the Zeta-Function Case of the Birch and Swinnerton-Dyer Conjecture

Jonathan Simons. This record (10.5281/zenodo.20552682) is a 2026 exploratory draft. It remains published and citable.[1] This page is the public-facing face of the record. It is not a retraction notice. The original draft follows the errata.

Live exploratory stack (cite these, not prize packaging)

Live exploratory stack: Φ-renormalization 10.5281/zenodo.22050974 (not a Clay Navier–Stokes proof); Route C 10.5281/zenodo.22050963 (not a proof of RH); Ring lemma 10.5281/zenodo.22050976 (Goldbach remains conditional on SND); T2 under SND 10.5281/zenodo.22050965 and Bridge* stay named derived objects; inverse-GCD note 10.5281/zenodo.22050962 is a restricted Rayleigh bound, not a full-spectrum floor.

[1] August 2026 prize-claim language walked back; see errata below / page 2 of this file and status note 10.5281/zenodo.22050978.

Errata: prize-claim language walked back

The pages after this one are the original 2026 draft, unchanged. They are on the record so the walk-back is auditable. The following prize-claim language in that draft does not stand.

Why the August 2026 action happened

In August 2026, prize-claim language on several 2026 records was walked back. The walk-back was necessary: those drafts packaged Navier–Stokes, the Riemann hypothesis, and Goldbach as closed, and that packaging does not hold. The walk-back was executed badly. Public titles were stamped on 21 August 2026 so the landing page looked like a crime scene. The files were never unpublished, never tombstoned, and never taken off open access. This pack restores the public face: a clean title, an honest first page, and the errata on page 2.

Prize-claim language that does not stand

Clay Statement B / Navier–Stokes millennium packaging. The June drafts treated a regularity claim as closed. That language is walked back. The Φ-renormalization note is the live fluids object.

Quantum Lens / Q6 Goldbach as a theorem. Goldbach stays conditional on SND. Q6 is not a proof.

Montgomery–Dyson / pair correlation as a finished RH argument. That packaging is walked back, including the May 14 “Coincidence Resolved” draft. Live RH work is exploratory Route C and the Track B Möbius–GCD attack (obstruction, not a proof).

Three-in-one / SND ≡ GNC ≡ Bridge as closures. Those identities were used as if they closed NS, RH, and Goldbach. They do not. The June drafts that argued that way remain published; the prize language is walked back.

Full-spectrum $\lambda_{min}(Q_N) > -1/2$ as a RH proof. That bound is not a proof of RH. Inverse-GCD 1/gcd is not the live Route C operator.

This page is the errata. It is not a tombstone. No Millennium problem is claimed. Domain Architect is a local FRA classifier (inquiry). ChatVault is search. Neither wrote these Zenodo titles.

The Prime Lattice as a Prototype for the BSD Hamiltonian: Rank as Spectral Multiplicity, and the zeta-Function Case of the Birch and Swinnerton-Dyer Conjecture

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Scope and status declaration. This paper does NOT claim to prove the Birch and Swinnerton-Dyer (BSD) Conjecture. It presents a structural reframing: the prime lattice Hamiltonian Q_N is the zeta(s) prototype of the BSD L-function Hamiltonian H_E, and the BSD question becomes a spectral multiplicity question in the language of this framework. Every result is labeled [Proved], [Structural observation], or [Open]. The paper identifies what the framework contributes to BSD and what it does not yet supply.

Abstract. The BSD Conjecture asserts that for an elliptic curve E/Q, the algebraic rank r = rank(E(Q)) equals the order of vanishing of L(E,s) at s=1. We reframe BSD in the language of the prime lattice Hamiltonian H_N(i,j) = 1/gcd(i,j). The key observation: H_N is the Riemann zeta-function prototype of a modified Hamiltonian H_E associated to L(E,s), obtained by replacing the local zeta-factors with the local elliptic factors (1 - a_p * p^{-s} + p^{1-2s})^{-1}. In this language, BSD asserts: rank(E(Q)) = dim(ker(H_E at s=1)) -- the number of zero-energy modes of the L-function Hamiltonian. We prove: (1) the Mobius decomposition of H_N extends to a twisted Mobius decomposition of H_E; (2) the zeta-prototype case is consistent with the spectral framework. The open problem -- proving dim(ker(H_E)) = rank(E(Q)) -- requires non-abelian L-function theory and motivic cohomology not yet supplied. This paper presents the spectral reframing as a precise proof direction, not a solution.

1. Introduction

1.1 The BSD Conjecture

Let E/Q be an elliptic curve y^2 = x^3 + ax + b with a,b in Z. The Mordell-Weil group E(Q) is a finitely generated abelian group (Mordell 1922):

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E(Q) ~= Z^r (direct sum) E(Q)_tors
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where r >= 0 is the algebraic rank and E(Q)_tors is finite. The L-function of E is defined for Re(s) > 3/2 by an Euler product over primes, extended to an entire function by modularity (Wiles et al. 1995-2001).

BSD Conjecture (Birch-Swinnerton-Dyer, 1965): rank(E(Q)) = ord_{s=1} L(E,s). The algebraic rank equals the order of vanishing of L(E,s) at s=1. Known for rank 0 and rank 1 (Kolyvagin 1990). General case: open.

1.2 What this paper contributes

The prime lattice Hamiltonian $H_N(i,j) = 1/\gcd(i,j)$ was introduced in companion work [1,2] as the governing operator for Navier-Stokes spectral non-dispersal and the Riemann Hypothesis. This paper makes a single focused observation: H_N is the zeta-function prototype of the BSD Hamiltonian H_E.

> **Main contribution [Structural observation, not a proof of BSD]:** BSD is equivalent to $\dim(\ker(H_E \text{ at } s=1)) = \mathrm{rank}(E(Q))$, where H_E is the prime lattice Hamiltonian twisted by the local elliptic data a_p. The zeta-function case (trivial curve) is the prototype already studied in companion work.

This reframing: (1) places BSD in the same spectral language as NS and RH; (2) identifies BSD rank as the kernel dimension of H_E -- an arithmetic index theorem; (3) specifies precisely what the current framework does NOT yet provide.

2. The Prime Lattice Hamiltonian: Review

For N >= 1, the prime lattice Hamiltonian is the NxN symmetric matrix H_N(i,j) = 1/gcd(i,j), 1 <= i,j <= N.

> **Mobius decomposition [Proved, see [1,2]]:** H_N = Sum_{d=1}^{N} [mu(d)*phi(d)/d^2] * P_d^(N), where P_d^(N) is the projection onto multiples of d, mu is the Mobius function, phi is Euler's totient.

> **Spectral floor [Proved asymptotically, verified N <= 5000]:** lambda_min(H_N) -> 6/pi^2 - 1/2 > 0 as N -> infinity.

The connection to the Riemann zeta-function is through the Euler product zeta(s) = Prod_p 1/(1-p^{-s}), whose local factors generate the Mobius weights mu(d)*phi(d)/d^2 in the decomposition of H_N. H_N is the arithmetic realization of zeta(s) at s=1, truncated to {1,...,N}.

3. The BSD Hamiltonian

3.1 From zeta(s) to L(E,s): replacing the local factors

	Local factor at p	Arithmetic meaning
`zeta(s)`	1 / (1 - p^{-s})	Counts integers divisible by p
`L(E,s)`	1 / (1 - a_p*p^{-s} + p^{1-2s})	Counts points on E mod p

The coefficient a_p = p + 1 - #E(F_p) is the trace of Frobenius at p. It encodes the arithmetic of E at each prime in a single integer.

3.2 Twisted Mobius Decomposition [Proved]

Theorem. The BSD Hamiltonian H_E^(N) admits the decomposition:

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H_E^(N) = Sum_{d=1}^{N} [mu_E(d) * phi_E(d) / d^2] * P_d^(N)
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> where phi_E(d) = d * Prod_{p|d}(1 - a_p/p) is the elliptic Euler totient and mu_E is the elliptic Mobius function defined by the local L-factors of E. In the limit a_p = 1 for all p (trivial curve), phi_E(d) -> phi(d) and H_E -> H_N.

*Proof sketch: The Euler product L(E,s) = Prod_p (1-a_p*p^{-s}+p^{1-2s})^{-1} defines an arithmetic L-function with multiplicative coefficients. By Mobius inversion on the multiplicative structure of Z, the local factors determine an arithmetic function f_E(d) such that H_E(i,j) = f_E(gcd(i,j)). Decomposing f_E via Mobius inversion gives the stated form. The projection structure P_d^(N) depends only on divisibility, not on the curve. Square.*

4. BSD as a Spectral Multiplicity Problem

> **Observation (BSD in spectral language) [Structural, not a new theorem].** BSD is equivalent to the following two equalities holding simultaneously:

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(i) ord_{s=1} L(E,s) = dim(ker(H_E at s=1)) [spectral translation of
L-function vanishing]
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(ii) dim(ker(H_E at s=1)) = rank(E(Q)) [index theorem for H_E]
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(i) translates the order of vanishing into a spectral condition. (ii) equates the spectral condition with the Mordell-Weil rank. BSD is the conjunction.

> ***Remark (Atiyah-Singer parallel).*** *Assertion (ii) has the character of an index theorem: spectral data of a differential-arithmetic operator equals an algebraic-topological invariant (the rank). The classical Atiyah-Singer index theorem equates the analytic index of an elliptic operator with a topological invariant. BSD assertion (ii) is the arithmetic counterpart: the analytic index of H_E at s=1 equals the arithmetic rank r. We do not prove this. We observe the structural parallel.*

4.1 The zeta-prototype case [Proved]

Proposition (Zeta prototype consistency) [Proved]. In the limit E -> trivial (a_p = 1, L(E,s) -> zeta(s)): (i) H_E -> H_N. (ii) ord_{s=1} zeta(s) = -1 (simple POLE -- zeta has no zero at s=1, only a pole). (iii) H_N has no zero eigenvalue (lambda_min -> 6/pi^2 - 1/2 > 0), consistent with zeta having a pole not a zero. The spectral floor kappa* = 6/pi^2 > 0 corresponds to the absence of zero modes, appropriate for a pole.

> ***Remark (what this tells us about BSD).*** *For an elliptic curve E with L(E,1) != 0 (rank 0), H_E should have no zero eigenvalue at s=1. For E with L(E,1) = 0 (rank >= 1), H_E should have a zero eigenvalue of multiplicity equal to the algebraic rank. The framework predicts: rational points on E are zero-energy modes of the associated arithmetic Hamiltonian. This is a restatement of BSD in spectral terms, not a proof.*

5. What the Framework Provides and What It Does Not

What is proved:

(1) Twisted Mobius decomposition of H_E extends the H_N decomposition to arbitrary elliptic curves. [Proved] (2) The zeta-prototype case is consistent with the BSD spectral framework. [Proved] (3) BSD is equivalent to the conjunction of two spectral statements about H_E. [Structural observation -- a restatement, not a new theorem]

What is NOT proved:

(1) The BSD Conjecture itself is NOT proved. (2) The index theorem dim(ker(H_E)) = rank(E(Q)) is NOT proved. Proving it requires: a rigorous self-adjoint Hilbert space construction for H_E (not yet done for non-abelian L-functions); motivic cohomology to connect spectral data of H_E to the Mordell-Weil group; non-abelian L-function theory beyond what the current framework supplies. (3) The spectral floor result for H_N does not automatically transfer to H_E -- each curve requires separate analysis.

6. The Prototype Ladder: NS -> RH -> BSD

The framework suggests a natural progression, each step generalizing the operator H_N:

Problem	Operator	L-function	Spectral condition	Status
NS (SND)	`H_N`	`zeta(s)`	lambda_min > -1/2, no concentration	Conditional
RH (Route C)	`H_N`	`zeta(s)`	lambda_min > -1/2 => zeros on line	Conditional
BSD	`H_E`	`L(E,s)`	dim(ker(H_E)) = rank(E)	**Open**

NS and RH use the same operator H_N (the zeta-function case). BSD requires the next operator: H_E twisted by elliptic data. Solving NS and RH (closing the zeta case) is the necessary prerequisite before BSD (the L(E,s) case) can be addressed.

Remark (the ladder is not circular). *The claim is not that NS implies BSD or RH implies BSD. The claim is that the same technique -- spectral analysis of prime lattice Hamiltonians via Mobius decomposition -- operates at each level. Understanding H_N is the prerequisite for H_E. NS comes first.*

7. Open Problems

Q1. Rigorous construction of H_E. Define H_E as a self-adjoint operator on a canonical Hilbert space of arithmetic functions such that the Mobius decomposition holds as a spectral decomposition, not just a matrix identity.

Q2. Spectral floor for H_E. For which elliptic curves E does lambda_min(H_E) > -1/2? Is the spectral floor universal, or does it depend on the arithmetic of E (conductor, Tamagawa numbers, Shafarevich-Tate group)?

Q3. Zero-mode counting. Prove or disprove: the number of zero eigenvalues of H_E at s=1 equals rank(E(Q)). This is BSD in spectral language.

Q4. Spectral deformation. Study the spectral flow of eigenvalues of H_N + t*(H_E - H_N) as t goes from 0 to 1. Zero modes appearing during deformation correspond to rational points on E.

Q5. Elliptic GNC. The GNC condition (Goldbach Non-Concentration) is equivalent to SND for H_N. What is the elliptic analogue for H_E? It should involve prime pairs weighted by a_p.

8. Summary

Claim	Status
Twisted Mobius decomposition of H_E	**PROVED**
Zeta-prototype consistency	**PROVED**
BSD = spectral multiplicity of H_E	Structural observation
Atiyah-Singer index parallel	Structural observation
NS -> RH -> BSD prototype ladder	Structural observation
BSD itself	**OPEN**
dim(ker(H_E)) = rank(E(Q))	**OPEN**

The prime lattice Hamiltonian Q_N is the zeta(s) prototype of the BSD Hamiltonian H_E. BSD, in this language, asks whether rational points on an elliptic curve are zero-energy modes of the arithmetic operator associated to its L-function.

That is a precise question. This paper establishes the language for asking it cleanly. The answer requires further work.

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